



Test Report # T16131-01-1 Issue 1
ANSI/ISEA Z89.1-2014 (R2019)
Plasticos del Golfo Sur S.A. de C.V
IGA Plagosur C – Industrial Safety Helmet
06 October 2021



Approved by:

A handwritten signature in black ink that reads "Sarah Henderson".

Sarah Henderson
Head Protection, Hand Protection &
Athletic Equipment - Laboratory Manager

Prepared by:

A handwritten signature in black ink that reads "Shelley Brady".

Shelley Brady
Head Protection & Athletic Equipment -
Laboratory Administrator

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Issued to: Plasticos del Golfo Sur S.A. de C.V.
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 Coatzacoalcos, Veracruz 96496
 Mexico

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Objective:

Contract testing to ANSI/ISEA Z89.1-2014 (R2019) “American National Standard for Industrial Head Protection”.

-Type I, Class E, G & C

Sample(s):

IGA Plagosur C - Industrial Safety Helmet

Description	Qty	Date Code	Material	Date Received	Sample ID
Orange Shell (Cap) w/ Ratchet Suspension (6 pt.), Non-vented	32	8/21	HDPE	08 September 2021	1A-x

Procedures:

Testing protocols in accord with good laboratory practice were employed unless otherwise specified, for all tests.

Testing procedures as specified within Section 10 of ANSI/ISEA Z89.1-2014 (R2019) were followed.

Testing was performed at room temperature 23°C ± 3°C (73.4°F ± 5.4°F) per ANSI/ISEA Z89.1 – 2014 (R2019).

Equipment:

Cadex Twin Wire Test Platform
 Environmental Conditioning Chambers
 Cadex Impact Software V6.9U
 Hanco Dielectric Strength Chamber

Assessment Summary:

Date Tested: 06 October 2021
 Test Performed By: Joe McGreal – Engineering Technician

Requirements	Compliant	Non-Compliant
6 Instruction and Marking	X	
7 Performance Requirements		
7.1 Requirements for Type I		
7.1.1 Flammability	X	
7.1.2 Force Transmission	X	
7.1.3 Apex Penetration	X	
7.1.4 Electrical Insulation Requirement		
7.1.4.3 Electrical Insulation Requirement - Class E	X	
7.3 Requirements for Optional Testing		
7.3.1 Reverse Wearing		
-Force Transmission		
7.3.2 High-Visibility		
7.3.3 Higher Temperature		
7.3.4 Lower Temperature		

Not Applicable

Samples as assessed meet the requirements of ANSI/ISEA Z89.1 2014 (R2019) for Type I, Class E, G & C.

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Results:

6	Instructions and Marking	Pass	Fail
	6.1 Instructions Provided	X	
	6.2 Marking:		
	-the name or mark of the manufacturer	X	
	-the date of manufacture	X	
	-the American National Standard designation (ANSI/ISEA Z89.1-2014)	X	
	-the applicable Type and Class followed by optional criteria markings (if applicable)	X	
	-the approximate headband size range	X	
	6.3 Optional Performance:		
	 – Reverse Donning LT – Lower Temperature HV – High Visibility HT – Higher Temperature	Not Applicable	

Note: All markings under clause 6.2 shall be at least 1.5 mm (0.06 in.) in height.

7.1.1 Flammability

Sample ID	Afterflame (sec)	Pass	Fail
1A-12	0	X	
Specification:	< 5		

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7.1.2 Force Transmission

Sample ID	Conditioning (°C)	Velocity (m/s)	Force (N)	Specified Value (N)	Observations	Pass	Fail
1A-1	49 ± 2	5.52	1656	≤ 4450	1	X	
1A-2		5.53	1701		1	X	
1A-3		5.51	1755		-	X	
1A-4		5.53	1720		-	X	
1A-5		5.52	1664		1	X	
1A-6		5.52	1664		1	X	
1A-7		5.53	1696		1	X	
1A-8		5.52	1782		-	X	
1A-9		5.53	1718		1	X	
1A-10		5.51	1637		1	X	
1A-11		5.53	1712		1	X	
1A-12		5.51	1735		-	X	
Average			1703	≤ 3780		X	
1A-13	-18 ± 2	5.52	2398	≤ 4450	-	X	
1A-14		5.50	2422		-	X	
1A-15		5.52	2333		-	X	
1A-16		5.52	2427		-	X	
1A-17		5.53	2383		-	X	
1A-18		5.53	2432		-	X	
1A-19		5.53	2503		-	X	
1A-20		5.51	2336		-	X	
1A-21		5.53	2487		-	X	
1A-22		5.53	2618		-	X	
1A-23		5.50	2490		-	X	
1A-24		5.53	2516		-	X	
Average			2445	≤ 3780		X	
Specification:		5.50 ± 0.05					

Observations:

(-) No observation

1 – Suspension tab released

7.1.3 Apex Penetration

Sample ID	Conditioning (°C)	Velocity (m/s)	Electrical Contact (Yes/ No)	Observations	Pass	Fail
1A-25	49 ± 2	6.9	No	-	X	
1A-26		7.0	No	-	X	
1A-27		7.0	No	-	X	
1A-28	-18 ± 2	7.0	No	-	X	
1A-29		7.0	No	-	X	
1A-30		7.0	No	-	X	
Specification:		7.0 ± 0.1	No Electrical Contact Allowed			

Observations:

(-) No observation

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7.1.4.3 Electrical Insulation Requirements (Class E)

Sample ID	Leakage (mA)	Burn Through (Yes/ No)	Observations	Pass	Fail
1A-1	4.4	No	-	X	
1A-13	4.6	No	-	X	
Specification:	Class E: ≤ 9 mA	No Burn Through Allowed			

Class E - 20,000 volts, 60 Hz, 3 minutes, 30,000 volts no burn through.

Note: Test line lowered to prevent flashover

Observations:

(-) No observation

Sample Photographs:



Observation Photograph:



Suspension tab released

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10. For the safety of our personnel, Client must advise if samples are known or suspected to contain hazardous substances. Safety Data Sheets must be provided upon request.
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12. Schedules are confirmed upon acceptance of quotation. All reasonable efforts will be made to comply with provided timeline. Guarantees are neither implied nor promised.
13. Certain work may be subcontracted to ICS-approved laboratories as required or applicable. Client will be notified of this in advance.
14. Client agrees to pay any and all additional costs associated with unexpected or above-standard communications and/or consultations with Client or third parties as designated by Client.
15. Client agrees to pay any and all additional costs for work additional to the original scope of work as agreed to by Client.
16. Client understands and agrees that ICS, in entering into this Contract and by performing services hereunder, does not assume, abridge, abrogate or undertake to discharge any duty or responsibility of Client to any other party or parties. No one other than Client shall have any right to rely on any Report or other representation or conduct of ICS and ICS disclaims any obligations of any nature whatsoever with respect to such third parties.
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18. Client agrees, in consideration of ICS undertaking to perform the test(s) hereunder, to protect, defend and indemnify ICS from any and all claims, damages, expenses either direct or consequential for injuries to persons or property arising out of or in consequence of the performance of the testing, inspection and reporting hereunder and/or the performance of the products tested or inspected hereunder, unless caused by the negligence of ICS.
19. It is agreed that if ICS should be found liable for any losses or damages attributable to the services hereunder in any respect, its liability shall not exceed the amount of the fee paid by Client for services rendered and Client's sole remedy at law or in equity shall be the right to recover that sum.
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21. ICS hereby objects to any conflicting terms contained in any order, acceptance or other subsequent correspondence submitted by Client.
22. In the event that payment is not received within 15 days of invoice date, Client agrees to pay a late payment charge on the unpaid balance equal to 1-1/2% per month or the maximum charge allowed by law, whichever is less, and all costs and expenses, including attorney's fees where recovery of the same is not prohibited by law, incurred by ICS in collecting such invoices.
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24. Cancelled/discontinued orders: Client responsible for all administrative and testing charges up to point of cancellation.